COMSATS University Islamabad, Lahore Campus
Department of Electrical and Computer Engineering

Assignment 2– SPRING 2026

Course Title:	Fundamentals of Digital Logic Design				Course Code:	EEE240	Credit Hours:	3(2,1)
Course Instructor:	Dr. Ali Raza				Programme Name:	BSE		
Semester:	3rd		SP25	Section:	A	Date:		
Submission Date:	**14th April, 2025**				**Maximum Marks:**		**40**	
Name:					**Registration Number:**			

<u>**Important Instructions / Guidelines:**</u>

- In case of plagiarism, you will be marked ZERO.

Problem 1:

Apply gate minimization technique using 4-variable K-map to simplify the following Boolean expression:

$$\bar{A}\bar{B}\bar{C}\bar{D} + A\bar{C}\bar{D} + \bar{B}C\bar{D} + \bar{A}BCD + B\bar{C}D$$

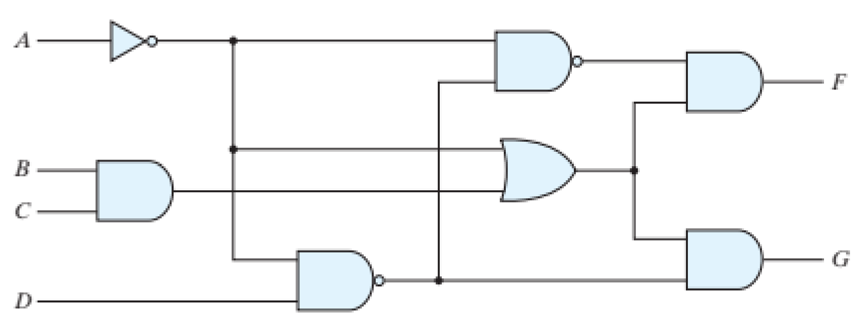
Problem 2:

Analyze the following Boolean function F, together with the don't-care conditions d, to express function in sum-of-minterms form:

$$F(A, B, C, D) = (0,6,8,13,14)$$
$$d(A, B, C, D) = (2,4,10)$$

Problem 3:

Using digital circuit analysis technique, evaluate the Boolean expression for output F and G in terms of the input variables A, B, C and D. Then use K-map to simplify the expression.



Problem 4:

Design a 4-bit combinational circuit that generates the 2's complement of a 4-bit binary input number. Provide the truth table, Karnaugh map (K-map), and the resulting circuit diagram in your working.